Lecture 32
1st Order Transients – 3 of 5

extensions and examples

RC circuit – $v_c(t)$

RL circuit – $i_L(t)$

} continuity

Other Stuff

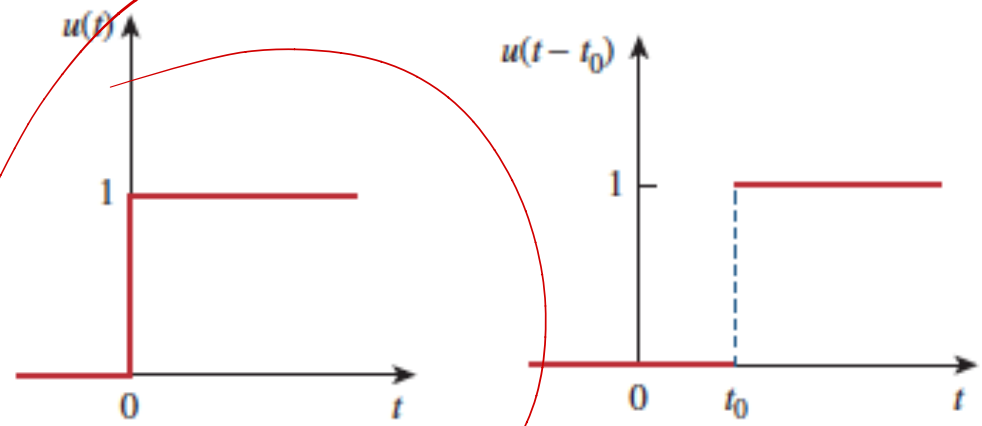
- Time invariance; for switch at $t = t_0$

$$x(t) = x(\infty) + (x(t_0) - x_\infty)e^{-(t-t_0)/\tau} + x_\infty$$

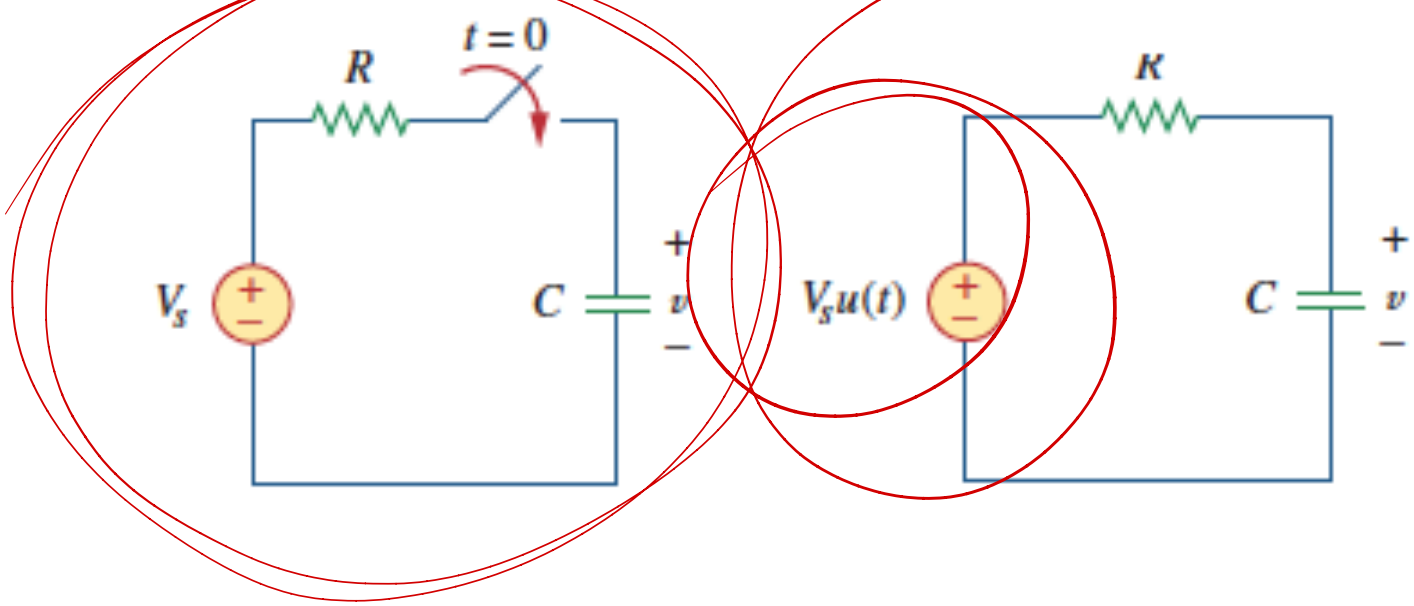
$t \geq t_0$

- Unit step function

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$



might replace the switch

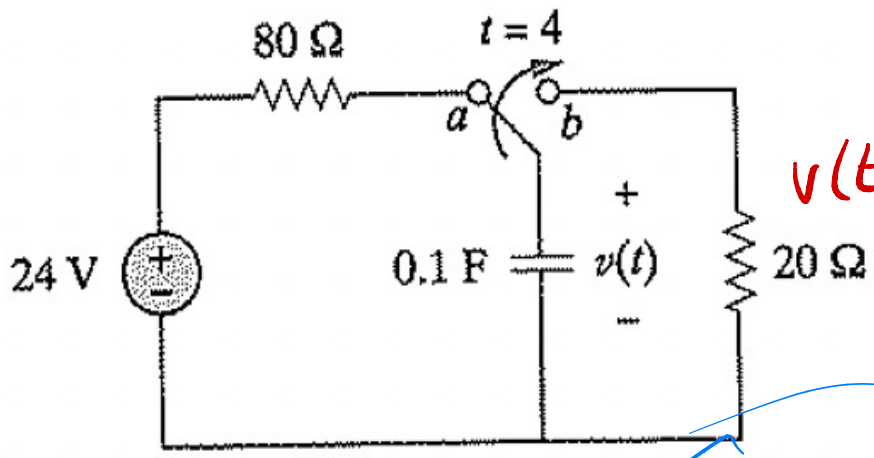


$$v(t) = 0 + 24e^{-(t-4)/2} \qquad t \geq 4$$

$$= 24e^{-t/2}e^{4/2}$$

$$= 24e^{4/2}e^{-t/2}$$

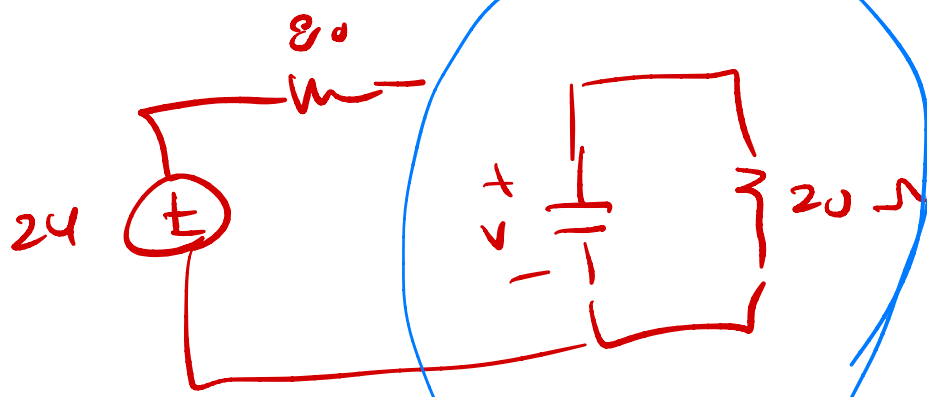
Example: note the 4 seconds



$$v(t) = V_\infty + (v(4) - V_\infty)e^{-\frac{(t-4)}{\tau}}$$

volts

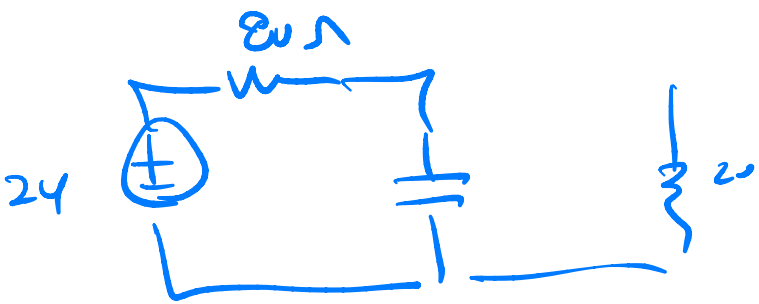
V_∞ :



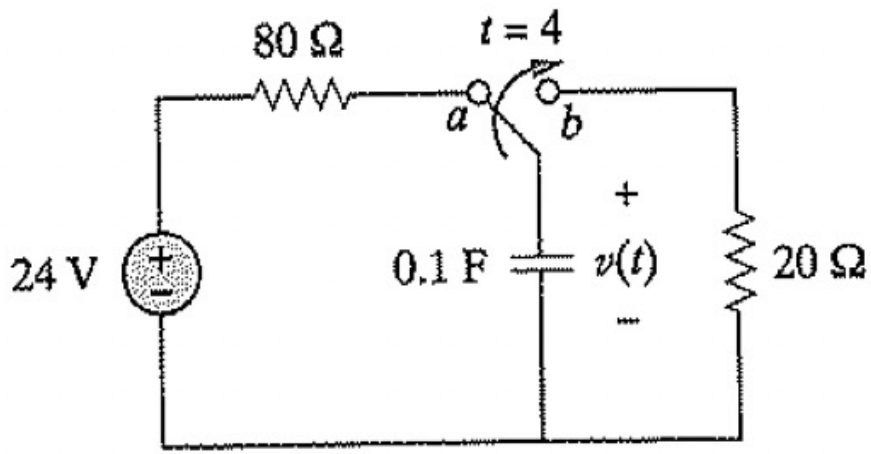
$V_\infty = 0$

$\tau = 20 \cdot (.1) = 2$ sec

$v(4)$:



$v(4) = 24$ volts



a→b:

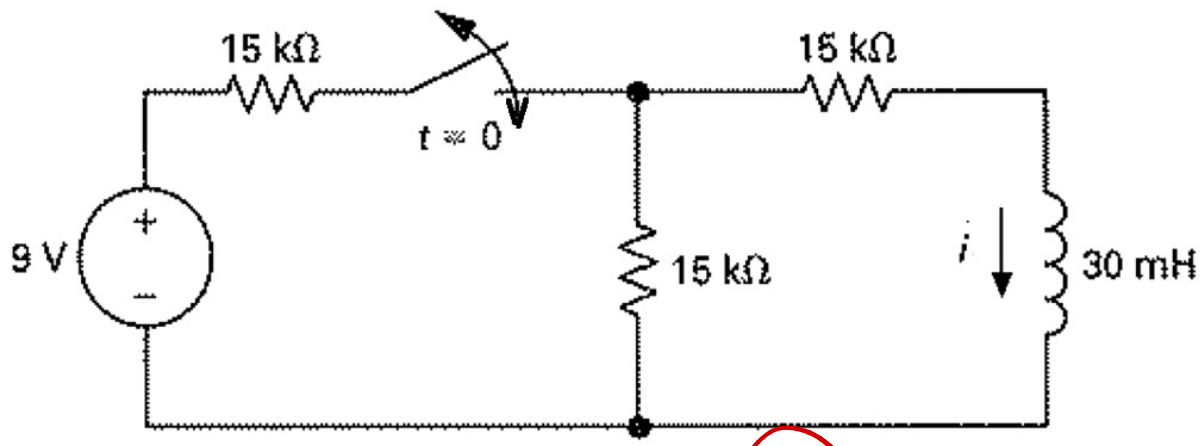
$$v(4) = 24\ V$$
$$v_\infty = 0\ V$$
$$R_{Th} = 20\ \Omega$$

b→a:

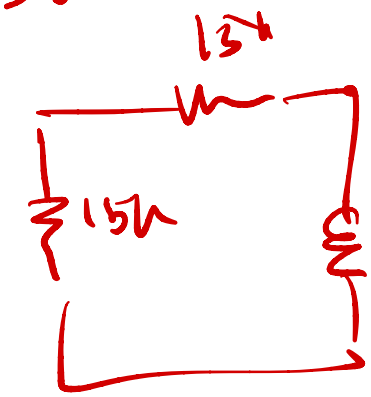
$$v(4) = 0\ V$$
$$v_\infty = 24\ V$$
$$R_{Th} = 80\ \Omega$$

$i(t) = i_\infty + (i_0 - i_\infty)e^{-t/\tau}$ $t>0$

Example: switch opens $= i_0$ $t<0$



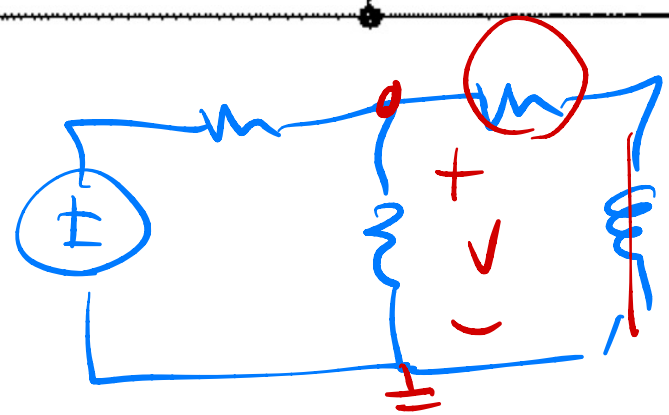
$t>0$



$i_\infty = 0$

$R_{Th} = 30k\Omega$

$t<0$



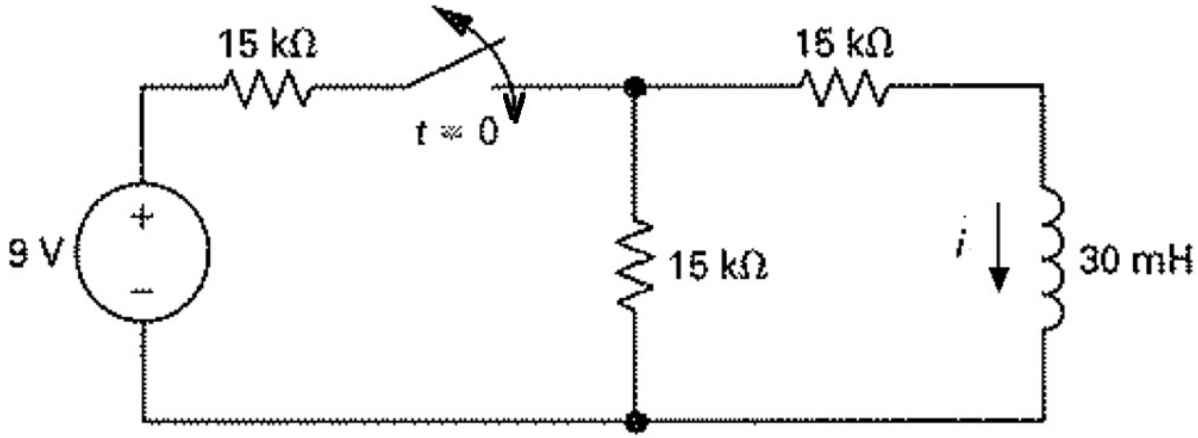
$i_0 =$

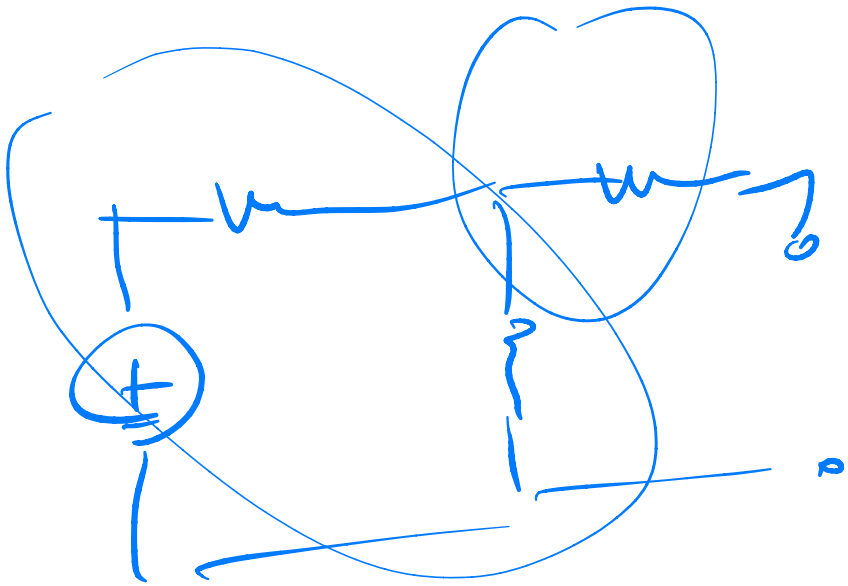
$$\frac{V-9}{15k} + \frac{V}{15k} + \frac{V}{15k} = 0$$

$$\frac{3V}{15k} = \frac{9}{15k}$$

$V = 3$ volts

$$i_0 = \frac{3}{15k} = 0.2\ mA$$





Opening at zero:

$i_0 = 0.2\ mA$

$i_\infty = 0\ mA$

$R_{Th} = 30\ k\Omega$

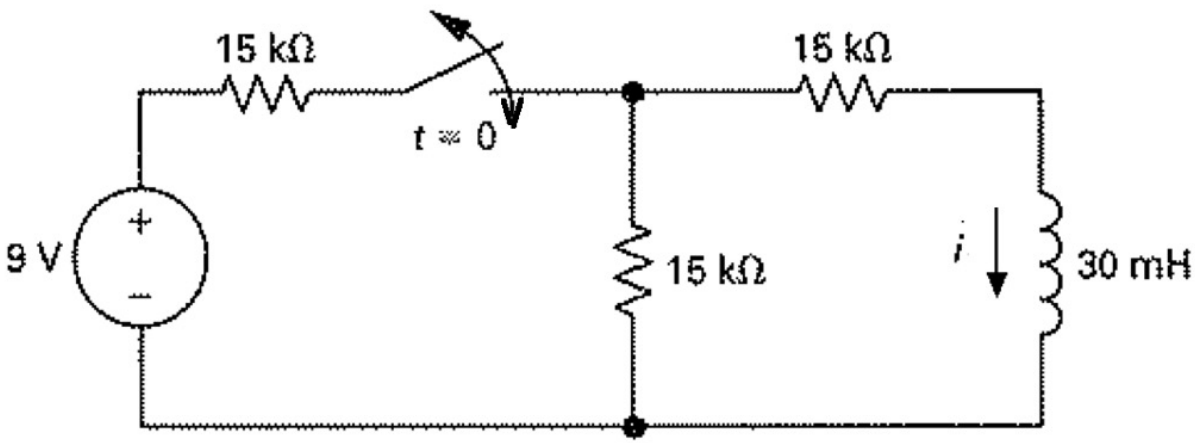
Example: switch closes

Closing at zero:

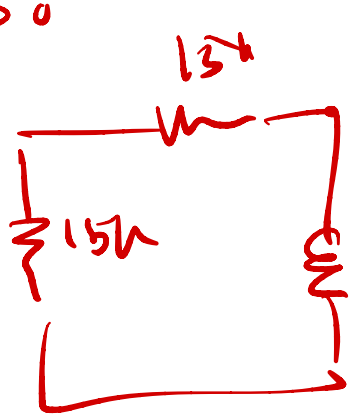
$i_0 = 0\ mA$

$i = 0.2\ mA$

$R_{Th} = 22.5\ k\Omega$

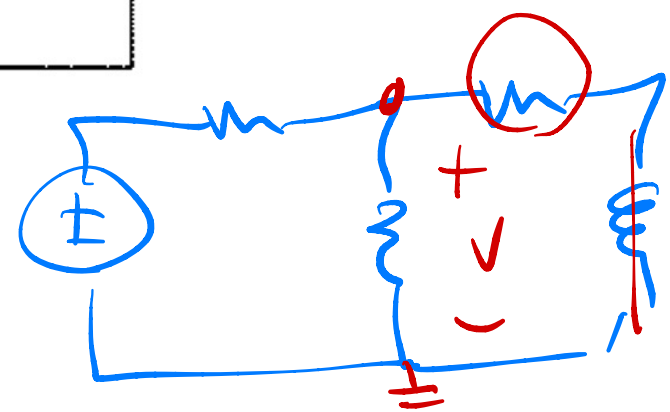


$t<0$ $t>0$



$i_0 = 0A$

$t>0$



$i_0 =$

$$\frac{V-9}{15k} + \frac{V}{15k} + \frac{V}{15k} = 0$$

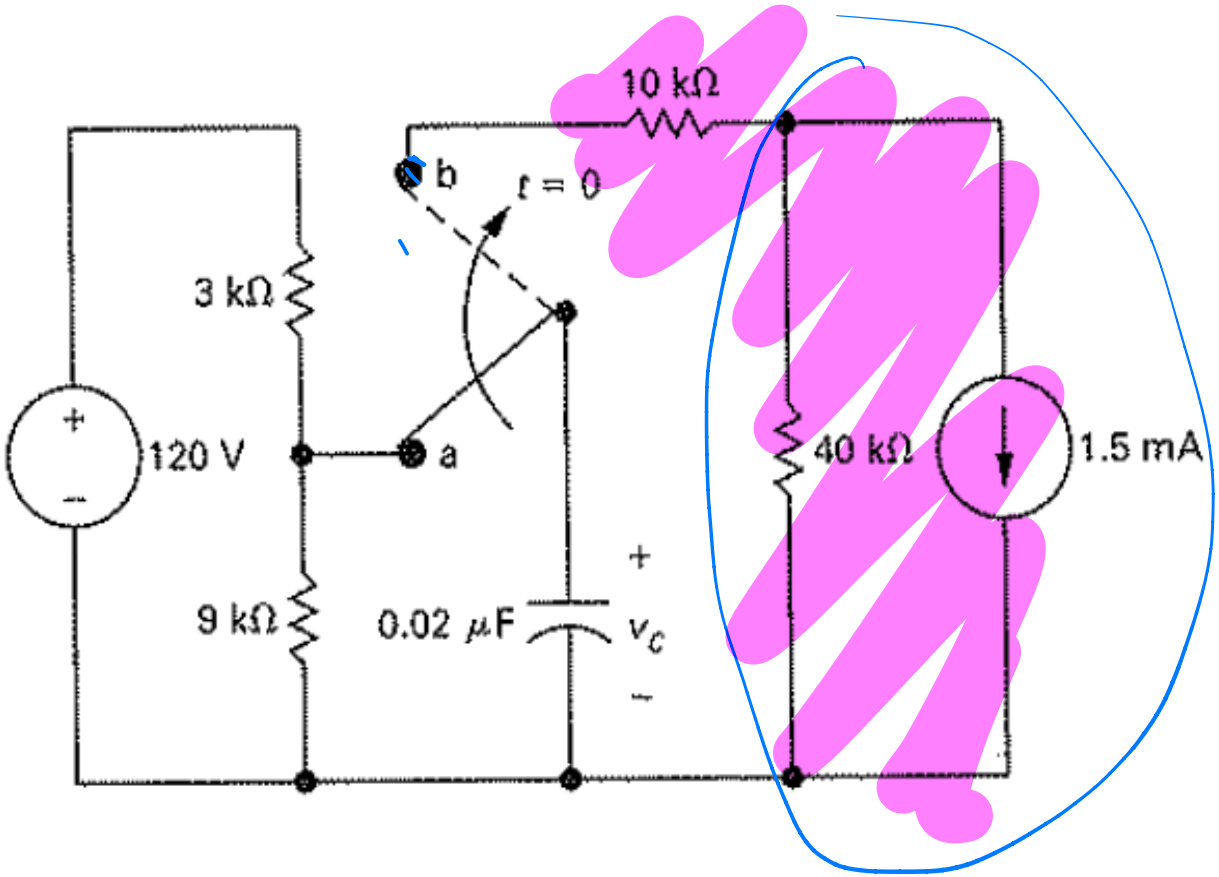
$R_{Th} = 22.5\ k\Omega$

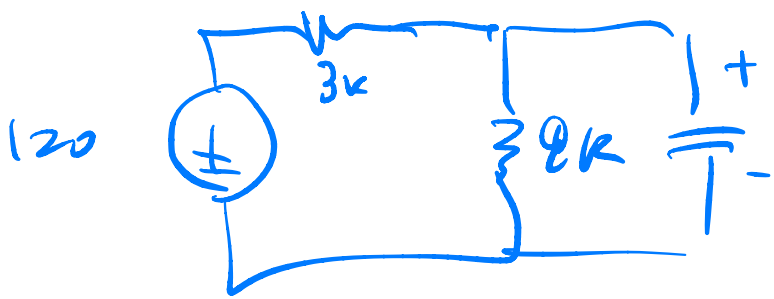
$$\frac{3V}{15k} = \frac{9}{15k}$$

$V = 3\ volts$

$$i_0 = \frac{3}{15k} = 0.2\ mA$$

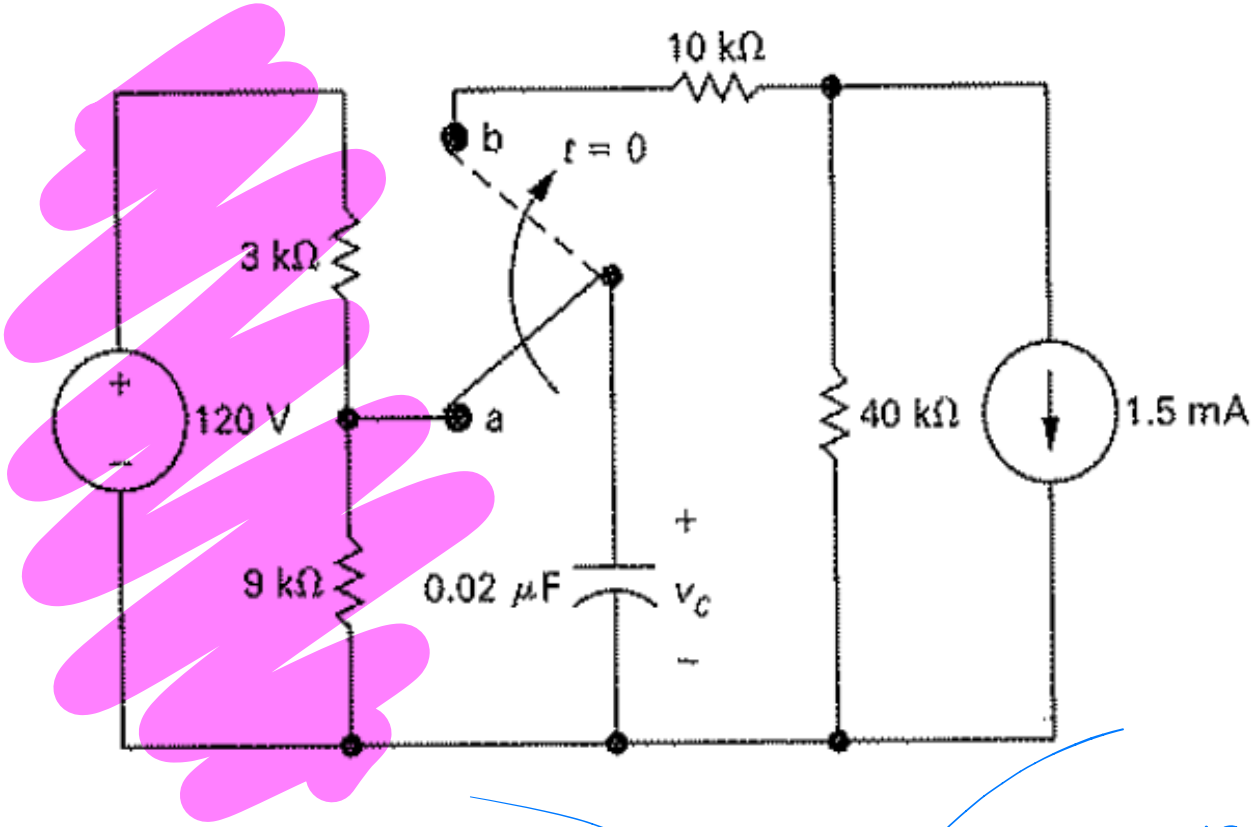
Example: switch moves from a to b



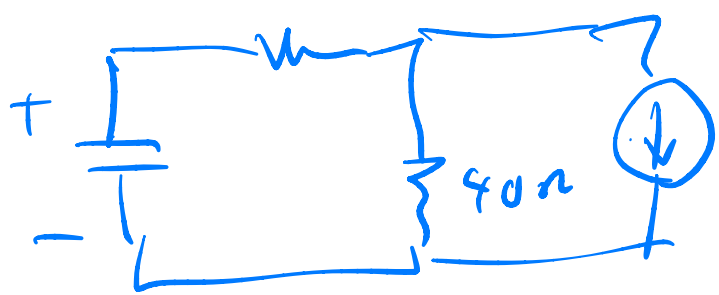


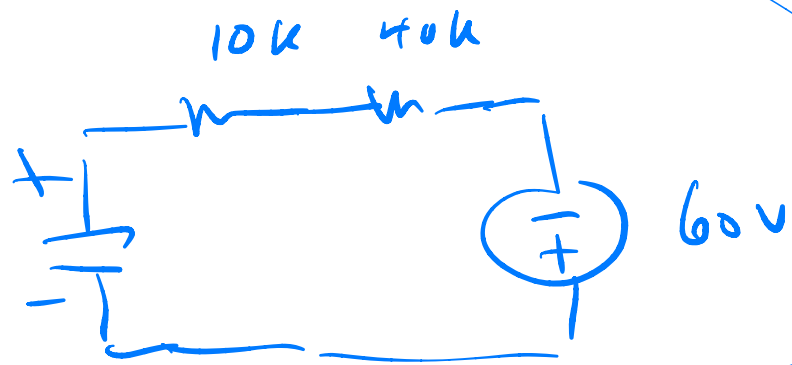
$V = 90$ v

$R_{th} = 3 \| 9 = \frac{9}{4}$ kΩ



$R_{Th} = 50k$

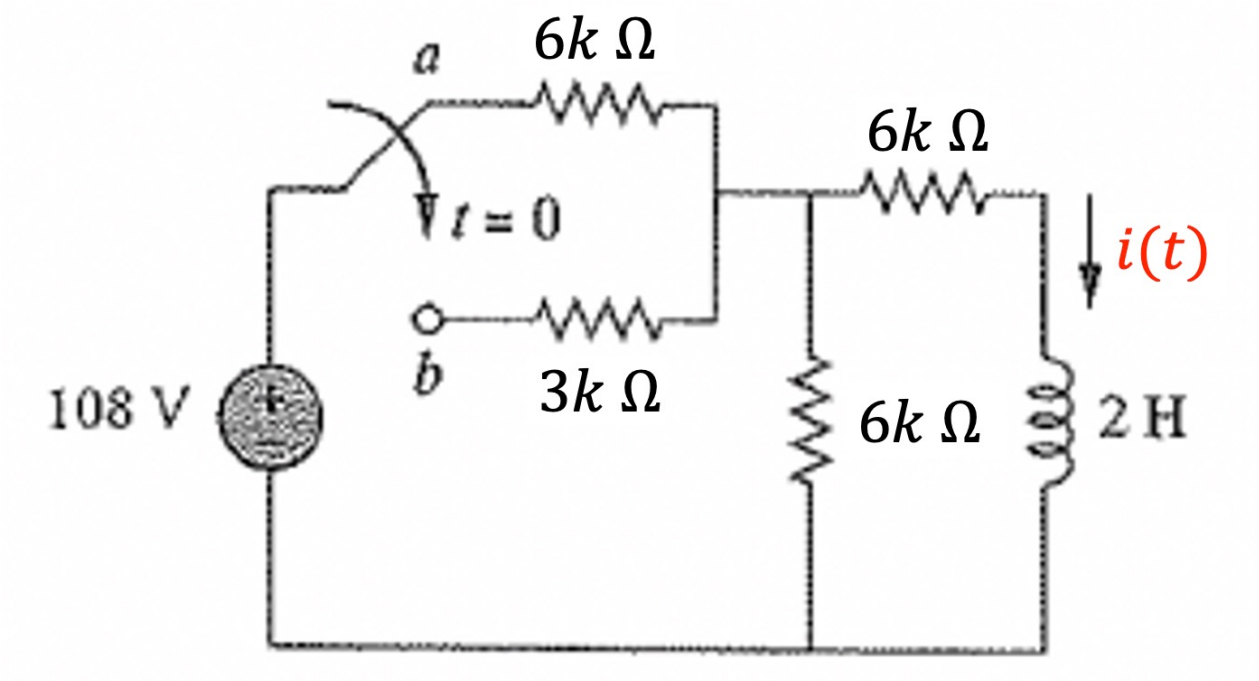


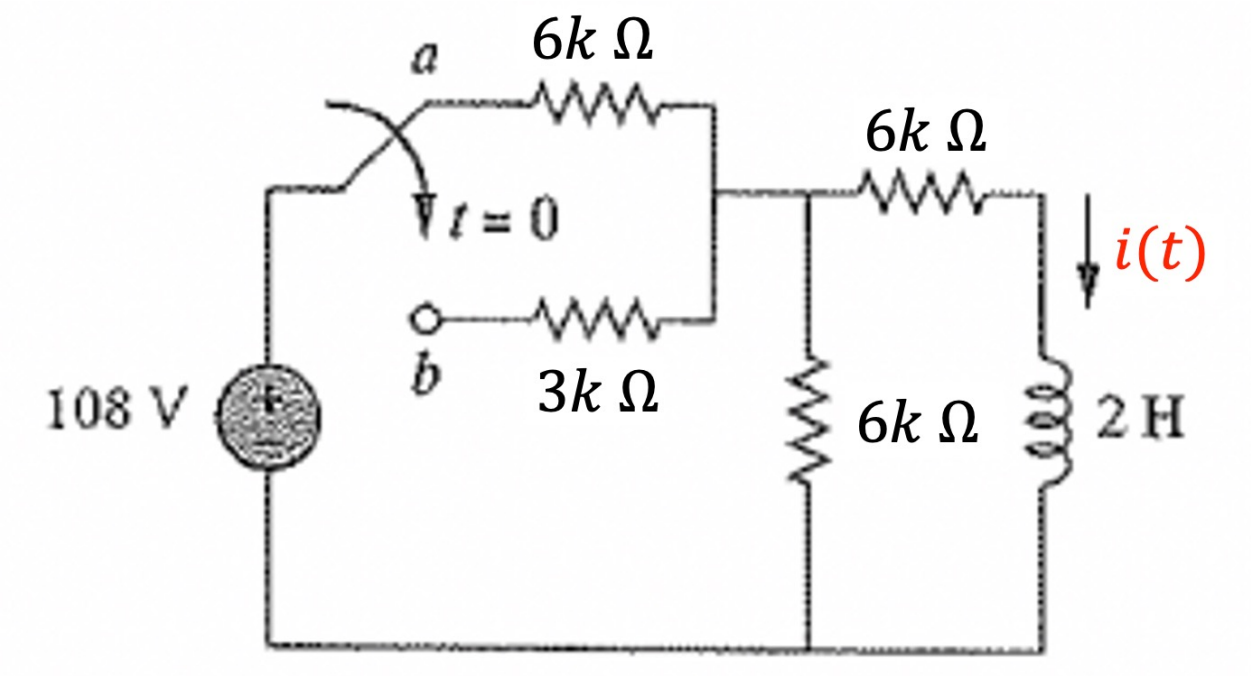


$V = -60$

$v_0 = 90\ V$
$v_\infty = -60\ V$
$R_{Th} = 50\ k\Omega$

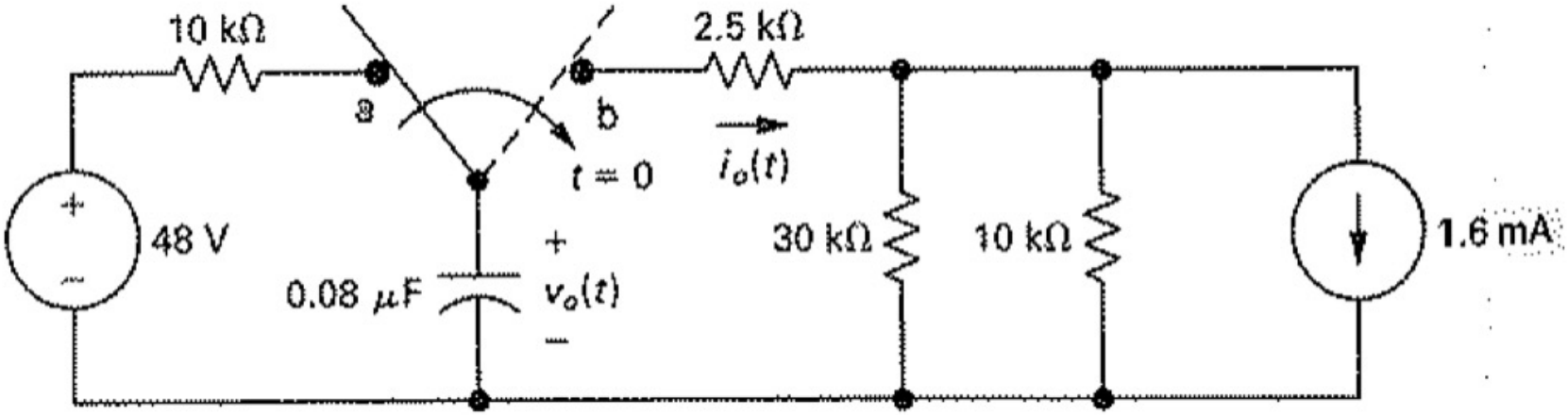
Practice problem: find $i(t)$

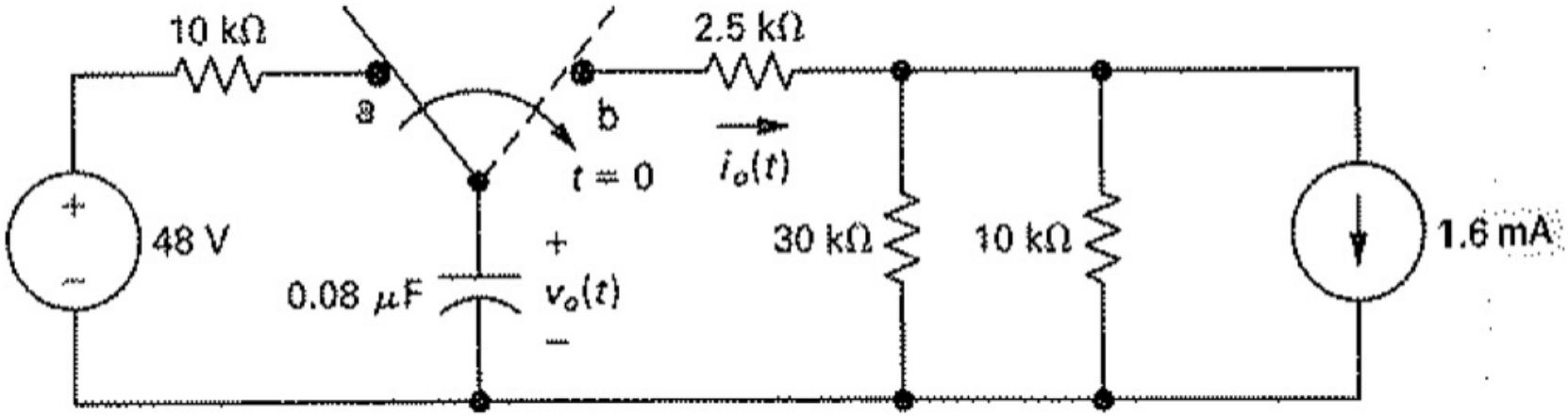




$$i(t) = -3\, e^{-4000t} + 9\ mA$$

Practice problem:

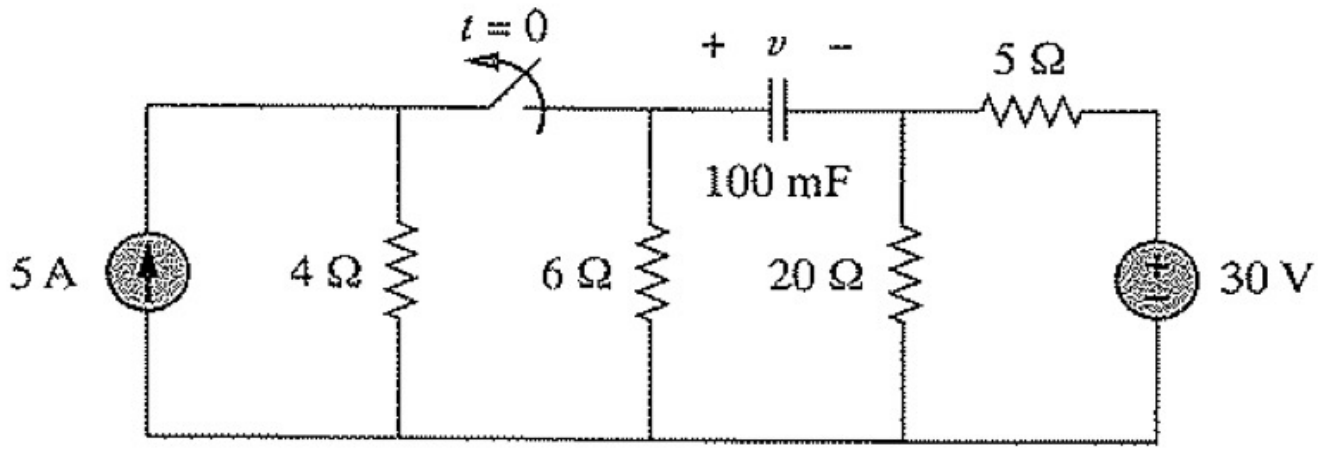




$$v(t) = 60\, e^{-125t} - 12\ V$$

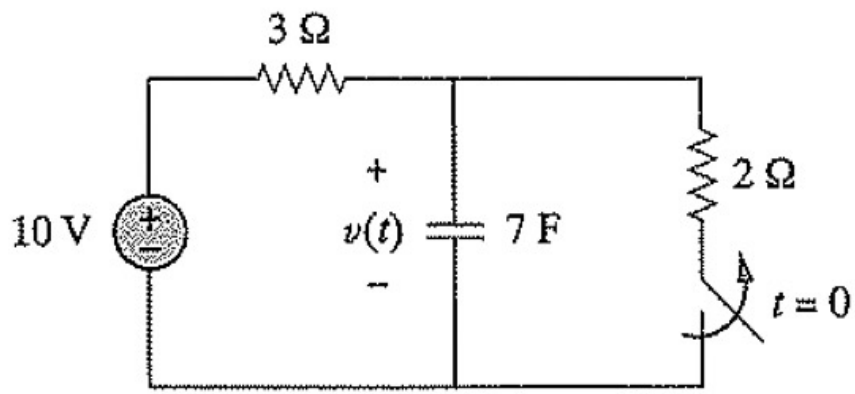
$v(t) = 12\, e^{-t} - 24\, V$

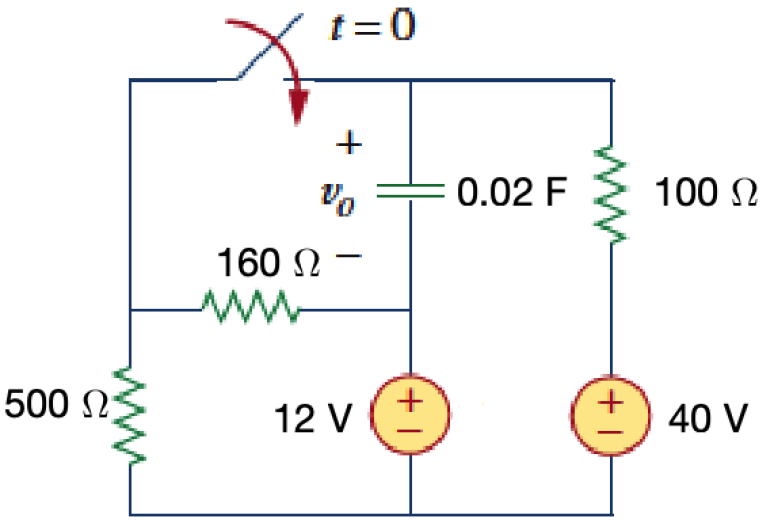
Practice problem:



$v(t) = -6e^{-t/21} + 10\ V$

Practice problem:



$v(0) = 28\ V$
$v_\infty = 14.0\ V$
$R = 54.8\ \Omega$

Practice problem: